

Development Methodology



27th January 2025

Note for the exam

Rules

This is an open-book exam: you are allowed to use handouts, notes, textbooks, and materials saved on your computer. However, if you use your computer for consultation, please ensure that airplane mode is activated.

Communication of any kind, whether among yourselves or with external parties, is strictly prohibited. Any violation of this rule will result in the invalidation of your exam.

Duration

1.5 hours.

How much should I write?

You should aim to write no more than 4 sides of paper. Use English and focus on including plenty of diagrams, charts, and visual aids, with minimal text. Prioritize clarity and conciseness to convey your ideas effectively. Use a black or blue pen; pencils or other colors are not allowed.

Pay attention

Write your name and student ID number at the top of the first page and clearly mark the sheet that represents the final version of your exam.

Any other questions?

Rise your hand if something is not clear, otherwise make (and write down) your assumptions.

Use case: Enerbrain (<https://www.enerbrain.com/>)

Introduction

Enerbrain is a company focused on reducing the energy consumption of buildings through IoT technologies and cloud solutions. They offer energy consulting services, environmental monitoring, energy monitoring, and optimization of HVAC systems (heating, ventilation, and air conditioning).

Objective

The objective of this case is to propose comprehensive architectural and organizational solutions to effectively implement Enerbrain's project.

Case Study Overview

Enerbrain's offerings include:

- Energy management systems for smart buildings
- Monitoring and visualization of energy consumption data
- Optimization of energy consumption and reduction of carbon emissions
- Use of machine learning algorithms to provide energy optimization strategies

Architectural Solutions

System Integration

Integrating various subsystems (environmental sensors, actuators, etc.) into a cohesive system is crucial. The architecture should allow seamless data flow between these components, ensuring real-time data collection and analysis.

Data Management

Implement a robust data management framework that includes:

- Data acquisition from sensors and building management systems (BMS) – you can assume that kind of items have already been available
- Data storage and processing using cloud solutions
- Real-time data access for customers through mobile and web applications

Artificial Intelligence (AI)

Develop AI models to analyze the collected data and provide useful insights. The AI should support:

- Predictive alerts for energy management
- Optimization of energy consumption (even through a conversational service)

Security and Privacy

Ensure data security and privacy by implementing encryption, secure access controls, and compliance with relevant regulations.

Organizational Solutions

Project Management

Adopt the methodology you deem best for this project.

Team Structure

Create a multidisciplinary team.